Chau Ngoc Thao Hien

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Ho Chi Minh City, Vietnam

PROFILE

Dedicated Biotechnology graduate with a profound passion for Pharmaceutical Sciences. With a proven research background in natural product extraction, *in vitro* bioactivity screening assays, and drug-protein interactions, I have developed hands-on expertise and rigorous analytical skills essential for scientific inquiry. My immediate objective is to pursue a PhD to master advanced methodologies, paving the way for a career as an independent researcher and a key contributor to pharmaceutical breakthroughs.

EDUCATION

- **2022 - present**
  International University - Vietnam National University Ho Chi Minh City
  Master of Science in Biotechnology
  - GPA: 89.5/100 (3.58/4.0)
- **2016 - 2020**
  International University - Vietnam National University Ho Chi Minh City
  Bachelor of Science in Biotechnology
  - GPA: 75.7/100 (3.07/4.0)

PROFESSIONAL EXPERIENCE

- **Mar 2022 - present**
  Teaching assisstant | International University - Vietnam National University Ho Chi Cinh City
  Courses: Analytical Chemistry, Microbiology, Pharmaceutical Biotechnology
  - Preparing experimental materials, chemicals, and equipment for practical classes
  - Guiding undergraduate students in performing experiments and analyzing results
  - Monitoring students during practical sessions to ensure compliance with laboratory safety regulations.
  - Tracking attendance and assissting with class administration
  - Grading quizzes, assignments, reports, and examinations
- **Jun 2019 - Sep 2019**
  Laboratory technician | Internship | Department of Medical Genetics - Tu Du Hospital
  Technique: Non-invasive Prenatal Testing (NIPT)
  - Extracting cell-free fetal DNA from maternal blood samples
  - Preparing DNA library and conducting PCR
  - Measuring relative quantity of DNA fragments by electrophoresis
  - Analyzing DNA by Next-generation sequencing (NGS)

RESEARCH INTERESTS

- **Medicinal Chemistry:** Isolation, purification, and characterization of bioactive secondary metabolites.
- **Pharmacology & Toxicology:** *In vitro* bioactivity screening and molecular mechanisms of action.
- **Drug-protein interactions:** Investigation of binding affinities, structural relationships, and the impact of drug-induced protein modifications.
- **Protein Biotechnology:** Recombinant protein expression, purification, and protein engineering.
- **Pharmaceutical Development:** Optimization of drug discovery pipelines.

RESEARCH EXPERIENCE

- **Apr 2023 - Apr 2026**

Master's thesis | International University - Vietnam National University Ho Chi Cinh City

Thesis title: "Investigating phytochemicals and biological activities of *Docynia indica* fruits collected in Kon Tum towards standardizing for drug development"

Thesis supervisor: Associate Professor Nguyen Hoang Khue Tu

- Fractioning medicinial plant extract using organic solvents and separating compounds in fractions by TLC
- Analyzing phytochemicals in plant extract by UPLC-MS/MS and finding their biological characteristics
- Assessing and studying mechanisms of action of antibacterial, antioxidant, anti-lipid peroxidation, and anti-tyrosinase activities using in-depth assays
- Predicting interactions between identified ligands and human CYP1A2 protein via *in silico* molecular docking

- **Feb 2023 - Feb 2024**

Student reseach project | International University - Vietnam National University Ho Chi Cinh City

Project title: **"**Volatile compounds, total phenolic contents with antimicrobial, antioxidant, tyrosinase inhibition and cytotoxicity of *Docynia indica* fruit extracts"

Project supervisor: Associate Professor Nguyen Hoang Khue Tu

Role: Principal investigator

- Obtaining plant extracts using different extraction methods and analyzing volatile organic compounds by GC-MS
- Conducting antimicrobial, antioxidant, and tyrosinase inhibitory assays
- Assessing cytotoxicity of plant extracts on mouse fibroblasts using MTT assay

- **May 2020 - Sep 2020**

Bachelor's thesis | International University - Vietnam National University Ho Chi Cinh City

Thesis title: **"**Volatile compounds, total phenolic contents with antimicrobial, antioxidant and anti-tyrosinase activities of *Docynia indica* fruit extracts"

Thesis supervisor: Associate Professor Nguyen Hoang Khue Tu

- Analyzing volatile organic compounds in plant extracts by GC-MS
- Quantifying polyphenol content in plant extracts and detecting specific phenolic compounds by TLC
- Determining antioxidant, antibacterial, and anti-tyrosinase activities of plant extracts using screening assays

PUBLICATIONS

- **Chau, H.**, & Nguyen, T. (2024). Antimicrobial activities of *Docynia indica* fruit extracts. World Journal of Pharmacy and Pharmaceutical Sciences, 13(5), 1-10.
- **Chau, N. T. H.**, Nguyen, D. P., Truong, H. A. V., & Nguyen, H. K. T.. Volatile organic compounds and anti-tyrosinase activity of *Docynia indica* fruit extracts. Journal of Analytical Sciences. *[Submitted manuscript]*

CONFERENCE PRESENTATIONS

- **Chau, N. T. H.**, & Nguyen, H. K. T. (2023, November). Composition, polyphenol content, antimicrobial and antioxidant activity of *Docynia indica* fruit extract. Fourth Scientific Conference for Students 2023, International University - VNU HCMC, Ho Chi Minh City, Vietnam. *[Poster presentation]*

HONORS AND AWARDS

- **2018** **Encouragement Scholarship | International University - Vietnam National University Ho Chi Cinh City**

RELEVANT AND ADDITIONAL SKILLS

- **Wet lab:** Bacterial cultivation, *in vitro* bioassays, UV-Vis spectroscopy, TLC, DNA extraction, DNA electrophoresis, protein expression and purification, SDS-PAGE, Western blotting
- **Dry lab:** Molecular docking, GraphPad Prism
- **Others:** Scientific writing, critical thinking and analysis

LANGUAGES

- Vietnamese (Native)
- English (IELTS 6.0)

ACADEMIC REFEREES

1. **Assoc. Prof. Nguyen Hoang Khue Tu**
   Associate Professor, School of Biotechnology
   International University - Vietnam National University Ho Chi Minh City
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